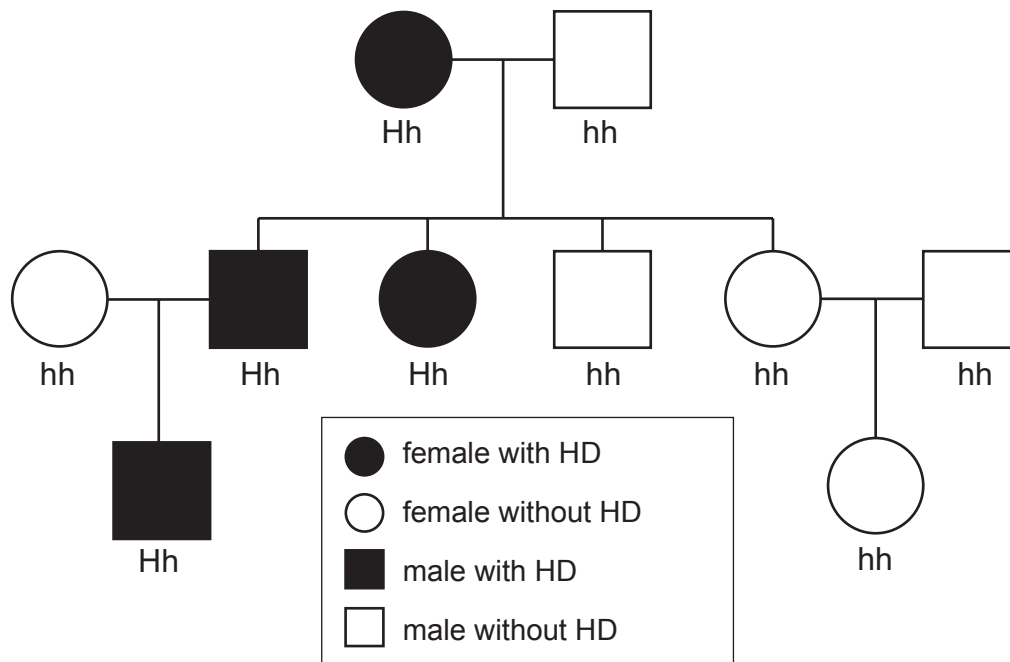


2. Huntington's disease (HD) is an inherited disease. People are born with the defective gene, but symptoms usually don't appear until middle age. Early symptoms of HD may include uncontrolled movements, clumsiness or balance problems. The disease is caused by an inherited defect in a single gene. With only one copy of the defective allele, a person can develop the disease.

The diagram shows a family tree identifying those who are unaffected (do not have HD) and those with HD.



- (a) (i) State the letter of the dominant allele. [1]
- (ii) State the genotype of a person with HD. [1]
- (b) **Complete** the Punnett squares and state the percentage chances of a child developing HD if:
- one parent does not have HD and one parent is heterozygous
 - both parents are heterozygous
- [4]

	H	h
h		
h		

% chance =

	H	h
H		
h		

% chance =

- (c) State **one** ethical problem faced by a person if they find out they have HD as a result of going for a genetic screening test. [1]

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